LecSeg-30 A Reproducible Low-Resource Benchmark & Diagnostic Study

Lecture-Video Topic Segmentation — Supervisor Reference Sheet

The Problem

Students cannot navigate 3-hour lecture recordings without chapter markers. **Goal:** automatically partition a lecture transcript into coherent topic segments. **Reference:** YouTube creator-supplied chapter timestamps (navigation metadata, not pedagogical gold standard).

Three notions of "topic boundary" (key conceptual distinction):

1 Discourse — rhetorical shift (definition→example). Detected by prosody, discourse markers. *Fine-grained.*

2 Semantic — embedding-space discontinuity. Detected by sentence transformers. *Medium-grained.*

3 Editorial — creator-placed chapter divider. LecSeg benchmark uses this. *Coarse-grained.*

Conflating levels 1 and 3 explains every acoustic signal failure.

Dataset — LecSeg-30

30 videos	5 domains (Bio, CS, Math, Phil, Phys)
32.52 h	total transcribed audio
46,525	sentences (spaCy)
419	YouTube chapter boundaries
904	reviewed subtopic labels
$\kappa_{chapter} = 0.535$	(inflated: shared YouTube metadata)
$\kappa_{subtopic} = 0.426$	(workflow consistency, not independent IAA)

Pipeline (end-to-end)

1. Download
yt-dlp → mp4; ffmpeg → 16kHz WAV

2. Transcription
faster-whisper large-v3 on vast.ai RTX 5090 ($3–4 total)

3. Sentence split
spaCy `en_core_web_sm`; merge <3-token fragments

4. Embed
BGE-large-en, E5-large, MPNet, MiniLM (CPU)

5. Multimodal features (optional)
CLIP frames · PaddleOCR slides · librosa prosody · TransNetV2 shots

6. Segment
11 methods (see table) → boundary indices

7. Evaluate
P_k / WD with bootstrap 95% CIs; Wilcoxon+Holm

Metrics

P_k **(Beeferman 1999)** and **WindowDiff (Pevzner 2002)** — window-based, lower = better. Slide a window of width $k = \lfloor N/2K \rfloor$ and count disagreements. Near-miss boundaries are not fully penalised — appropriate for the navigation task. $P_k = 0.5$ = random; $P_k = 0.0$ = perfect. F1/BS are secondary diagnostics only.

Methods Tested & Results

Method	$P_k \downarrow$	Verdict
TextTiling, KMeansSeg	0.605, 0.617	**Fail**
CosineSeg, BertSeg	0.489–0.490	**Weak**
C99	0.422	**Weak**
BGE-Divisive (baseline)	0.388	**Best baseline**
CLIP+Text fusion	0.374	**Good**
Cross-model (E5+BGE)	0.371	**Best P_k/WD**
Balanced Selector	**0.359**	**Best mean**

Four Pipeline Components (N1–N4)

Label	What it does
N1	Two-stage predictor: broad pass (high recall) → refinement (high precision)
N2⋆	Entropy-weighted fusion: $w(m) \propto e^{-H(m)}$; high-entropy (flat) signals auto-down-weighted
N3	Hierarchical enforcer: $\mathcal{B}_{chapter} \subseteq \mathcal{B}_{subtopic}$
N4	Llama-3.1-8B titling (local, offline); boundary shifting does not improve P_k

⋆N2 is the most distinctive; N1/N3/N4 are engineering adaptations.

Two Central Findings

Finding 1 — Oracle gap (bottleneck is selection, not generation)

Candidate pool $P_k = 0.017$, recall=96.8% → Selector $P_k = 0.359$ → Gap ≈ 0.34 all in ranking

Nearly perfect boundaries exist in the pool — we cannot always pick them. **Next step:** supervised boundary ranker (∼50–100 labelled lectures).

Finding 2 — Granularity mismatch (most transferable result)

fires at discourse level (too fine)
Prosody, BERTopic, disc. markers: P_k>0.41

fires at editorial level (right)
CLIP visual: $P_k = 0.396$ fused: $P_k = 0.374$

Acoustic signals are accurate at the *wrong* granularity. CLIP succeeds because slide transitions are themselves editorial decisions.

Statistical Evidence

Cross-model vs baseline: P_k, WD **significant** ($p < 0.05$, Wilcoxon+Holm). Selector vs cross-model: BS, F1 **significant**; P_k/WD **not significant**. Leave-domain-out: $P_k = 0.4012$ (selector is a local operating point, not universal).

Core claim: LecSeg-30 does not solve lecture segmentation — it makes it **measurable**, **inspectable**, and **reproducible**.